

Note – Do not use existing virtual machines for the following practical. Ensure date and time are visible.

1. Use appropriate VMs to fetch the available port numbers of the target system. Later install IIS on the target machine and see if port 80 is also present in the list.
2. Create an analysis report using appropriate website with SSL details for wipro.com and rpsconsulting.in.
3. Login to Azure account and list the names of the users logged-in in last 7 days.
4. Create a powershell script to execute (on DC or your personal machine) the following:
 - a. List hostname
 - b. List current logged-in username
 - c. List IP Address
 - d. Test connectivity from internet
 - e. Trace network path from your computer to google.com
 - f. List all open ports within your machine.
5. Download a docker container in a freshly installed CentOS 9 machine. Use the following command to pull the image from DockerHub (**[docker pull tomarj44/c7custom:v3](#)**).
6. Use telnet to capture the username and password using appropriate command/tools. Ensure the 2 linux VMs are in same lan segment (week3-demo) and also, they must be pinging each other.
7. In one of the linux server, add the following port numbers/services and validate using commands:
 - a. Add port 80 (with proper protocol)
 - b. Add port 53 (with proper protocol)
 - c. Add NTP service
 - d. Block ICMP ping request.
 - e. Block FTP ports
8. Install a new windows server and create a new domain controller machine with the forest name as “**wiprotest.org**” with a new member machine (name it “client2”) also, ensure to domain join it.
9. Perform the same actions (same as question 7) on domain controller machine and verify.
10. Add a 20GB HDD on this domain controller machine with NTFS file system and verify.
11. Import a fresh Palo-Alto VM and the policy to block the following websites
 - a. www.youtube.com
 - b. www.facebook.com
12. Use appropriate VMs to capture the “tcpdump” packets while access a file on port 8080. During this, ensure to capture the event ID 1116 and 1117 using proper tools.
13. Write proper powershell script to list top 10 event logs for event ID 4688 and verify it.
14. Edit existing powershell script and monitor the following application and it must generate alerts/logs when these applications are opened and verify.
 - a. Calc.exe
 - b. Notepad.exe
 - c. Cmd.exe
 - d. Powershell.exe
 - e. MSTSC
15. List top 10 application event logs from DC machine.
16. Write an AI prompt asking the ChatGPT/Gemini to list an example of phishing email to target yourself. Display the valid prompt and the response from the AI here.
17. Use Cisco Packet Tracer and configure DHCP server and verify if client machine is able to receive dynamic IP address from this DHCP server.
18. Use windows machine (client2) to connect securely to one of the linux machine using command prompt/powershell.
19. Fetch a PDF file using appropriate Google Dork from “Wipro.com”.
20. Use azure calculator to calculate the cost (in INR only) for the existing setup within your VMWare environment.